

TUSTIN MCAF, CA, USA

WMO: 722915

Lat: 33.700N

Lon: 117.833W

Elev: 17

StdP: 101.13

Time Zone: -8.00 (NAP)

Period: 86-97

WBAN: 93114

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	3.9	5.2	-8.6	1.8	14.0	-5.9	2.3	15.3	11.3	18.8	9.1	17.1	0.3	70	0.340

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.8	33.9	20.9	31.9	20.6	30.2	20.3	23.0	30.1	22.3	29.5	21.6	28.6	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	15.3	26.9	19.7	14.5	26.4	18.9	13.8	25.7	68.0	30.0	65.3	29.6	62.7	28.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	5.9	5.3	1.7	38.6	1.5	2.6	0.7	40.5	-0.2	42.1	-1.0	43.5	-2.1	45.5		
			-0.5	24.9	1.4	1.2	-1.5	25.8	-2.3	26.5	-3.1	27.2	-4.1	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	14.0	14.8	16.2	18.0	19.2	20.8	22.7	23.7	22.7	20.5	16.8	13.8
	DBStd	4.07	2.45	2.85	2.28	2.44	2.02	2.00	1.81	1.96	2.41	2.34	2.38	2.28
	HDD10.0	6	2	2	0	0	0	0	0	0	0	0	0	2
	HDD18.3	566	136	103	73	33	14	3	0	0	1	6	56	141
	CDD10.0	3153	125	136	193	241	286	325	394	424	381	324	204	120
	CDD18.3	672	1	4	7	24	41	78	136	166	132	72	11	1
	CDH23.3	5388	77	99	149	334	253	494	859	1261	1047	546	229	38
CDH26.7	1702	12	26	43	128	51	132	216	429	394	204	62	6	
Wind	WSAvg	1.8	1.5	1.7	1.7	2.1	2.2	2.2	2.2	2.0	1.7	1.7	1.5	1.4
Precipitation	PrecAvg	269	62	68	55	21	3	1	1	3	9	5	31	37
	PrecMax	663	200	316	253	103	12	12	5	44	49	26	134	106
	PrecMin	90	0	0	0	0	0	0	0	0	0	0	0	1
	PrecStd	149	58	76	58	28	3	2	1	9	14	6	31	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.3	29.9	30.9	34.9	30.7	34.0	33.1	35.0	36.4	36.4	32.0	27.1
		MCWB	14.2	16.1	17.5	18.2	20.6	20.6	22.1	22.1	21.7	19.6	17.5	14.3
	2%	DB	25.7	26.3	27.2	30.1	27.8	29.8	30.6	32.6	32.9	31.8	28.3	24.0
		MCWB	13.8	15.3	15.6	16.9	18.8	20.2	21.8	22.0	21.9	19.2	15.7	14.3
	5%	DB	23.0	23.4	24.4	27.1	26.0	27.8	28.9	30.8	30.7	28.3	25.9	21.9
		MCWB	13.2	13.8	15.6	17.2	18.0	19.7	21.0	21.7	21.5	19.3	15.1	13.8
	10%	DB	20.5	21.0	22.2	24.3	24.4	26.0	27.5	28.9	28.7	26.0	23.7	20.0
		MCWB	12.7	13.4	15.5	16.8	17.7	18.9	20.5	21.2	20.9	18.4	15.1	13.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	19.1	19.0	20.0	21.0	22.0	23.7	24.1	24.0	22.5	19.7	17.6
		MCDB	22.7	27.1	25.6	27.9	28.0	30.2	30.2	31.8	32.1	29.9	25.3	22.9
	2%	WB	16.1	16.7	17.7	18.9	19.7	21.0	22.5	23.0	22.8	21.2	18.5	16.2
		MCDB	19.4	22.5	24.1	26.9	26.1	28.5	29.0	30.2	30.2	28.2	24.2	20.4
	5%	WB	15.1	15.6	16.7	18.0	18.7	20.1	21.6	22.3	21.9	20.2	17.4	15.2
		MCDB	19.2	19.9	22.6	25.4	24.8	26.9	27.8	29.5	29.2	26.2	23.1	19.1
	10%	WB	14.2	14.7	15.8	17.1	17.9	19.3	20.8	21.5	21.1	19.3	16.5	14.2
		MCDB	18.6	18.8	21.1	23.7	23.6	25.5	26.7	28.3	28.2	24.7	21.7	18.5
Mean Daily Temperature Range	5% DB	MDBR	13.0	11.9	11.7	12.0	10.1	10.1	9.8	10.8	11.8	12.1	14.0	13.0
		MCDBR	18.8	17.4	17.3	18.6	13.5	13.9	12.0	13.6	15.7	17.4	18.9	17.1
		MCWBR	10.2	9.4	9.5	9.5	7.0	7.0	5.9	6.7	7.5	9.0	9.9	10.1
	5% WB	MCDBR	12.4	12.4	13.5	14.0	11.0	12.0	10.8	11.9	13.4	13.4	14.1	13.2
		MCWBR	8.3	7.9	8.4	8.1	6.1	6.5	5.8	6.3	6.9	7.6	8.9	9.2
Clear-Sky Solar Irradiance	taub	0.323	0.332	0.356	0.363	0.361	0.376	0.383	0.367	0.353	0.353	0.311	0.310	
	taud	2.536	2.523	2.430	2.418	2.457	2.416	2.467	2.495	2.526	2.500	2.610	2.613	
	Ebn at Noon	893	921	919	924	924	905	897	908	911	884	903	892	
	Edh at Noon	82	92	109	116	113	117	111	106	98	93	76	73	
All-Sky Solar Radiation	RadAvg	3.21	4.09	5.46	6.63	7.10	7.47	7.45	7.09	5.99	4.55	3.54	2.90	
	RadStd	0.22	0.40	0.29	0.34	0.48	0.46	0.39	0.20	0.24	0.33	0.21	0.21	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
Regional (50 neighbors)		+0.63	N/A	-1.70	+0.72	N/A	N/A	N/A	-112	+223

Nomenclature: See separate page